

Event-Driven ML Pipeline - Real-Time Stream Processing

Maharshi Soni | real-time-ai | advanced

Description

Event-driven stream processing for real-time ML inference. Async message broker, windowed aggregation, real-time features, online prediction. No external broker needed.

Tech Stack

fastapi, uvicorn, scikit-learn, numpy, pandas, pydantic, pytest

Features

- Async pub/sub broker
- Tumbling and sliding windows
- Real-time feature engineering
- Online prediction
- Dead letter queue
- Monitoring dashboard

Sample Input

Event streams

```
python main.py --train && curl localhost:8000/events
```

Sample Output

Real-time predictions

```
{"event_id": "e_123", "prediction": "anomaly", "confidence": 0.91}
```

Project Files

```
.gitignore
main.py
project-idea.json
pyproject.toml
README.md
requirements.txt
.github/workflows/test.yml
models/model.joblib
models/scaler.joblib
src/config.py
```

```
src/synthetic.py
src/__init__.py
src/api/dashboard.py
src/api/__init__.py
src/pipeline/broker.py
src/pipeline/dead_letter.py
src/pipeline/events.py
src/pipeline/features.py
src/pipeline/predictor.py
src/pipeline/replay.py
src/pipeline/windows.py
src/pipeline/__init__.py
tests/conftest.py
tests/test_api.py
tests/test_broker.py
tests/test_dead_letter.py
tests/test_events.py
tests/test_features.py
tests/test_predictor.py
tests/test_replay.py
tests/test_windows.py
tests/__init__.py
```

Test Results

PASSED

Run: 27 | Passed: 27 | Failed: 0